

A Multi-Stream Temporal Context Model for Narrative Memory

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Abstract

The Temporal Context Model⁹ and the Context Maintenance and Retrieval model that extends it¹⁵ together provide a widely-used computational framework for understanding how memory for a sequence of events depends on the temporal context in which they were encoded. Cornell and Zhang⁵ recently extended CMR with a slower list-level *hierarchical context* that synchronizes to the item-level context at list boundaries and unifies within-list and between-list organization in free recall. Here we introduce a *Multi-Stream Temporal Context Model* (MS-TCM), a generalization of Cornell and Zhang’s hierarchical CMR for memory of *interleaved storylines*. MS-TCM augments hierarchical CMR with three new mechanisms: (i) a bank of *per-storyline* contexts that each drift only while their own storyline is active; (ii) *storyline-return reinstatement at encoding* with strength λ , so that returning to a previously-abandoned storyline partially restores its cached context rather than forcing it to drift through intervening other-storyline events; and (iii) a *storyline-initiation mixture* at recall onset with weight τ , in which the first recall is, with probability τ , cued by the storyline hierarchy rather than by the end-of-list item context. We are also developing a fourth extension — strict hierarchical recall with a separate global cross-storyline context (Iteration 2, implementation in progress) — whose architecture we describe alongside the committed Iteration 1. We derive analytically why the λ mechanism predicts the equivalence of cued recall across narratively-adjacent events that are temporally adjacent (*grouped*) versus separated by 2–7 other-

storyline events (*bridge*), an equivalence reported by Xu et al.¹⁸ with Bayes factor $BF_{01} = 412$. We then fit MS-TCM (Iteration 1) to the category condition of Manning et al.¹²'s public free-recall dataset by maximum likelihood, compare it to a verified, equation-derived implementation of Cornell and Zhang hierarchical CMR, and report two behavioral diagnostics that motivate Iteration 2: the early-vs-late lag-CRP difference is a category-composition shift rather than a within-condition mechanism difference, and the late-list serial position scallop is fully accounted for ($r = 0.999$) by category-organized recall. MS-TCM achieves a log-likelihood improvement of $\Delta LL = +197.66$ over hierarchical CMR ($\Delta AIC = -391.32$) on the FRFR-category data, with $\hat{\tau} = 0.408$.

1 The puzzle of memory across narrative storylines

Recency and contiguity in free recall. Two robust principles organize human memory for a sequence of events. The first is *recency*: items encountered late in a sequence are remembered better than items encountered earlier. The second is *contiguity*: successful recall of one item biases the next recall toward items that were encoded close to it in time^{9,11,14}. Both principles vary smoothly with the temporal separation between events. In free recall of word lists, contiguity is most clearly visible in the conditional response probability as a function of lag (lag-CRP), which peaks at lag +1 (the immediately-following item) and decays for larger lags¹⁰. The same principles show up at much longer time scales: in continuous-distractor free recall, a recency-like decay unfolds over tens of seconds^{3,8}.

The *Temporal Context Model* (TCM;⁹) offers a simple explanation for both principles. A single d -dimensional temporal context vector $\mathbf{c}(t)$ is maintained across encoding. Each encoded item f_i at time t updates the context as

$$\mathbf{c}(t) = \rho \mathbf{c}(t-1) + \beta \mathbf{c}_i^{\text{IN}}, \quad (1)$$

where $\rho = \sqrt{1 - \beta^2}$ preserves $\|\mathbf{c}(t)\| = 1$ under unit-norm inputs and $\beta \in (0, 1)$ is the drift rate. The context similarity $\mathbf{c}(t) \cdot \mathbf{c}(t')$ decays as $\rho^{|t-t'|}$, so items encoded near each other in time share more context than items encoded far apart; at retrieval, reinstatement of a cue's encoding context

preferentially activates its temporal neighbours. TCM produces both recency (end-of-list items dominate retrieval because their context overlaps most with the retrieval cue) and contiguity (nearby items share context and are cross-cued). The *Context Maintenance and Retrieval* model (CMR;¹⁵) extends this machinery with a second sub-region of context carrying source or task information, and jointly captures semantic, temporal, and source clustering. Cornell and Zhang⁵ extend CMR with a *slower* hierarchical list-level context that drifts at rate β_{list} (alongside the item-level β_{enc}) and synchronizes to the item-level context at list boundaries, unifying within-list and between-list organization within a single memory-search model. Our model takes Cornell and Zhang’s hierarchical CMR as its theoretical baseline; we treat the published Cornell–Zhang equations as the reference implementation against which our extensions are compared.

Memory for interleaved storylines. For decades the cluster of phenomena around TCM and CMR was studied almost exclusively with experimenter-generated sequences of words. Over the last ten years, an increasingly rich literature has begun to ask how the same contextual principles apply when the sequence is a naturalistic narrative – a television show, a movie, or a spoken story – that humans watch or listen to in a continuous stream^{1,7,20}. Two distinctive features of naturalistic narratives are relevant here.

First, *narratives are structured by events*. Event-segmentation theory¹⁹ and the situation-models literature²¹ argue that ongoing experience is parsed into discrete event representations at boundaries marked by changes in characters, setting, goals, or causal structure. Event boundaries modulate memory: items encoded across a boundary are remembered as further apart in time than items within the same event, even when objective time is matched^{6,7}. Shared schema structure within an event supports robust reinstatement after interruption².

Second, *naturalistic narratives often interleave multiple storylines*. Many television shows and novels tell their stories by cutting between parallel threads: the viewer experiences, moment to moment, a single linear stream of events that is the braided-together interleaving of multiple ongoing storylines. *Why Women Kill* is one such show; *This Is Us* is another. A viewer who watches such a show experiences events from storyline *A*, then events from storyline *B*, then events from

74 storyline *A* again, and so on, even though within each storyline events would follow one another
75 continuously if unspooled. Figure 1 illustrates the metaphor: each storyline is a strand with its
76 own internal timeline, and the viewer’s experience is a braid that splices segments from the strands
77 together.

78 **The empirical puzzle.** Xu et al.¹⁸ asked whether cued recall of narrative events obeys the same
79 temporal-contiguity principles that govern cued recall of word lists. Their participants watched
80 the first season of *Why Women Kill* – a show that interleaves three storylines set in different historical
81 periods – and were later cued with one event from the show and asked to recall the event that
82 preceded or followed it. The cued-to-target pairs fell into three conditions:

- 83 1. **Within-event** (Experiment 2): cue and target belong to the same event (same storyline, same
84 scene).
- 85 2. **Across-event-within-storyline** (Experiment 1, “grouped”): cue and target are consecutive
86 events on the same storyline with no intervening events from other storylines.
- 87 3. **Across-event-bridge** (Experiment 3, “interleaved”): cue and target are narratively consecu-
88 tive on the same storyline but separated by $m = 2\text{--}7$ intervening events from other storylines
89 (mean ≈ 3).

90 A single-stream contiguity account predicts a substantial decrement for the bridge condition
91 relative to the grouped condition: the bridge target is several times further away in the viewer’s
92 temporal context than the grouped target, and under Equation 1 context similarity falls as ρ^{m+1} . In
93 the data, by contrast, the bridge and grouped conditions are statistically indistinguishable, with a
94 Bayes factor $\text{BF}_{01} = 412$ in favour of the null hypothesis of no difference between them. This is the
95 puzzle we seek to explain.

2 The Multi-Stream Temporal Context Model

MS-TCM is built on top of Cornell and Zhang⁵'s hierarchical CMR. We have implemented the Cornell and Zhang⁵ model directly from its published equations, verified the implementation to machine epsilon against hand-derived oracles (see `code/ms_tcm/_likelihood_core.py` and `code/tests/test_hcmr_*.py`), and use it as the comparator against which MS-TCM's contribution is measured. Iteration 1 of MS-TCM (committed; `code/ms_tcm/_likelihood_core_mstcm.py`) augments that baseline with three mechanisms: *per-storyline contexts*, the λ *storyline-return reinforcement at encoding* (v6 Eq. 6; `notes/two_level_cmr_v6.pdf`), and the τ *storyline-initiation mixture* at recall onset. Iteration 2 of MS-TCM (in progress; `code/ms_tcm/_likelihood_core_mstcm_v2.py`) restructures the architecture into a strict two-level hierarchy with a separate global cross-storyline context and per-storyline associative matrices; the empirical fit of Iteration 2 is pending. The architectural schematic for the committed Iteration 1 is shown in Figure 2.

2.1 Item-level and storyline-level contexts

At any given encoding step i on active storyline $s(i) \in \{1, \dots, K\}$, MS-TCM maintains an *item-level* context $\mathbf{c}^{\text{item}}(i)$ that drifts at every step with rate β_{enc} , and a bank of *storyline-level* contexts $\{\mathbf{c}_k^{\text{story}}(i)\}_{k=1}^K$, of which only the active one $\mathbf{c}_{s(i)}^{\text{story}}$ updates on step i . The inactive storyline contexts are frozen until their own storyline resumes:

$$\mathbf{c}^{\text{item}}(i) = \rho_{\text{enc}} \mathbf{c}^{\text{item}}(i-1) + \beta_{\text{enc}} \mathbf{c}_i^{\text{IN}}, \quad (2)$$

$$\mathbf{c}_{s(i)}^{\text{story}}(i) = \rho_{\text{story}} \mathbf{c}_{s(i)}^{\text{story}}(i-1) + \beta_{\text{story}} \mathbf{c}_i^{\text{IN}}, \quad (3)$$

$$\mathbf{c}_k^{\text{story}}(i) = \mathbf{c}_k^{\text{story}}(i-1) \quad \text{for all } k \neq s(i), \quad (4)$$

with $\rho_{\text{enc}} = \sqrt{1 - \beta_{\text{enc}}^2}$ and $\rho_{\text{story}} = \sqrt{1 - \beta_{\text{story}}^2}$ preserving unit norm under unit-norm inputs.

Cornell and Zhang⁵'s hierarchical CMR has a single list-level context $\mathbf{c}^{\text{list},G}$ that drifts at every step with rate β_{list} ; MS-TCM replaces this with K per-storyline contexts, one per active storyline, each drifting only when its own storyline is encoded. When the experiment exposes only one storyline

117 ($K = 1$), MS-TCM’s per-storyline context recovers Cornell and Zhang’s list-level context exactly.

118 **Feature inputs.** Each item i enters the context update through an input vector $\mathbf{c}_i^{\text{IN}}$ that mixes a
 119 pre-experimental component with an experimentally-accumulated component (see §2.5):

$$\mathbf{c}_i^{\text{IN}} = (1 - \gamma_{\text{fc}}) M_{\text{pre}}^{\text{FC}} f_i + \gamma_{\text{fc}} M_{\text{exp}}^{\text{FC}} f_i. \quad (5)$$

120 In the worked example in §4 the pre-experimental matrix $M_{\text{pre}}^{\text{FC}}$ is the identity (orthogonal one-hot
 121 items), so $M_{\text{pre}}^{\text{FC}} f_i$ is just f_i ; γ_{fc} is still identifiable through the Hebbian-accumulated $M_{\text{exp}}^{\text{FC}}$.

122 2.2 Storyline boundary mechanisms

123 Three boundary transitions shape the storyline-context dynamics. At a *within-storyline event bound-*
 124 *ary* ($e(i) \neq e(i-1)$, $s(i) = s(i-1)$), the item-level context synchronizes to the active storyline’s context:
 125 $\mathbf{c}^{\text{item}}(i) \leftarrow \mathbf{c}_{s(i)}^{\text{story}}$. At a *storyline switch* ($s(i) \neq s(i-1)$), the outgoing storyline’s context is cached into
 126 a storyline-context associative matrix M^{SC} via the Hebbian update

$$\Delta M^{\text{SC}} = g_{s(i-1)} \mathbf{c}_{s(i-1)}^{\text{story}} (i-1)^\top, \quad (6)$$

127 where g_s is a one-hot indicator of storyline s in the storyline index space. The item-level context
 128 then synchronizes to the incoming storyline’s context. At a *storyline return* — the case that motivates
 129 the λ mechanism — the returning storyline’s context is partially reinstated from M^{SC} with strength
 130 $\lambda \in [0, 1]$:

$$\mathbf{c}_{s(i)}^{\text{story}}(i) \leftarrow \lambda (M^{\text{SC}\top} g_{s(i)}) + (1 - \lambda) \mathbf{c}_{s(i)}^{\text{story}}(i-1). \quad (7)$$

131 At $\lambda = 0$, Equation 7 recovers Cornell and Zhang hierarchical CMR’s per-storyline counterpart with
 132 no return cache. At $\lambda = 1$, the context is fully restored to its cached snapshot. Intermediate values
 133 blend the two. After Equation 7 fires, the item-level context synchronizes to the reinstated $\mathbf{c}_{s(i)}^{\text{story}}(i)$.

134 The item-to-context associative matrix M^{IC} accumulates Hebbian updates at every encoding step:

$$\Delta M^{\text{IC}} = \mathbf{c}^{\text{item}}(i) f_i^\top. \quad (8)$$

135 2.3 Retrieval: a τ -mixture over initiation routes

136 Cornell and Zhang hierarchical CMR initiates each recall list from an end-of-encoding item-level
 137 context (recency-driven), with a hierarchical fallback to the list-level beginning-of-list context
 138 only after item-level retrieval fails. Empirically, however, the FRFR-category data show a strong
 139 primacy spike in the probability of first recall ($\hat{p}(\text{first recall} \mid \text{sp} = 1) \approx 0.27$) that the recency-driven
 140 initiation cannot reproduce (§4). MS-TCM therefore adds a τ *mixture* at recall onset that promotes
 141 the storyline-cued initiation route from a fallback to a first-class option.

142 With probability $1 - \tau$, the recall list is initiated by the standard end-of-encoding item-level cue.
 143 With probability τ , a storyline $\hat{s}$ is sampled from the storyline hierarchy via

$$P(\hat{s} = s \mid \mathbf{c}^{\text{item}}(W)) = \frac{\exp(k (M^{\text{SC}} \mathbf{c}^{\text{item}}(W))_s)}{\sum_{s'} \exp(k (M^{\text{SC}} \mathbf{c}^{\text{item}}(W))_{s'})}, \quad (9)$$

144 where $\mathbf{c}^{\text{item}}(W)$ is the item-level context at the end of encoding step W , and the item-level retrieval
 145 cue is reactivated to $\mathbf{c}_s^{\text{story}}$ before sampling the first recall. After the first recall, both initiation
 146 routes converge onto the standard Cornell and Zhang dynamics: the cue context drifts toward
 147 the last-recalled item's input context at rate β_{rec} , and the within-trial stopping probability $p_{\text{stop}} =$
 148 $\exp(-\varepsilon_d a^{\text{nr}}/a^{\text{r}})$ governs when recall terminates⁵ C&Z 2025 Eqs. 3, 7. At every retrieval step,
 149 candidate items j are activated through

$$a_j = M^{\text{IC}\top} \mathbf{c}^{\text{ret}}, \quad (10)$$

150 and the probability of recalling item j is

$$P(j | \mathbf{c}^{\text{ret}}) = \frac{\exp(ka_j)}{\sum_{\ell \in C} \exp(ka_\ell)}, \quad (11)$$

151 where $k > 0$ is the softmax gain and C is the set of still-in-play candidate items (standard CMR
152 no-repeats convention). Setting $\tau = 0$ and $K = 1$ recovers the Cornell and Zhang retrieval rule
153 exactly.

154 2.4 Iteration 2: strict hierarchical recall with a global cross-storyline context

155 Behavioral diagnostics on the FRFR-category data (§4.1) reveal that participants organize their
156 recall by category even when the presentation order is randomized – a signature that the Iteration
157 1 architecture only partly produces. We are therefore developing Iteration 2, which restructures
158 MS-TCM into a strict two-level hierarchy. *Per-storyline* variables ($\mathbf{c}_s^{\text{story}}$, plus a per-storyline list-
159 level context, plus per-storyline associative matrices $M_{\text{exp}_s}^{\text{FC}}$ and M_s^{CF}) drift only when their own
160 storyline is active; a separate *global* cross-storyline context $\mathbf{c}^{\text{global}}$ drifts at every encoding step at
161 rate β_{enc}^G , with global associative matrices $M_{\text{exp}_G}^{\text{FC}}$, M_G^{CF} , and a M_G^{lists} matrix indexed by storyline that
162 holds each storyline’s end-of-storyline context. Recall under Iteration 2 is a τ -mixture over two
163 routes:

- 164 1. **Route α (probability $1 - \tau$, recency-driven).** The end-of-encoding global context $\mathbf{c}^{\text{global}}(W)$
165 cues items through M_G^{CF} , with no storyline lookup. This is a flat, global-only retrieval
166 analogous to standard CMR readout.
- 167 2. **Route β (probability τ , strict hierarchical).** The global context cues a storyline, $\hat{s} \sim \text{softmax}(k M_G^{\text{lists}} \mathbf{c}^{\text{global}})$;
168 the within-storyline cue is set to the storyline’s beginning-of-list context (producing within-
169 storyline primacy); items are recalled from M_s^{CF} until the storyline-stop fires; the procedure
170 returns to the global level with $\hat{s}$ excluded from the next candidate set (one-step exclusion).
171 This produces the category-organized recall sequences observed in the data.

172 Iteration 2 retains the λ storyline-return reinstatement at encoding (Eq. 7). When $K = 1$,

Iteration 2 reduces to the Cornell and Zhang two-level hierarchy with τ -controlled hierarchical-fallback weight; when $\tau = 0$, Iteration 2 reduces to a flat global-context recall analogous to standard CMR. The empirical fit of Iteration 2 to FRFR-category is in progress.

2.5 The pre-experimental context matrix

The mixture weight γ_{fc} in Equation 5 balances two kinds of input context: a *pre-experimental* component $M_{pre}^{FC} f_i$ that reflects what the learner already knew about item i 's semantic neighborhood before the experiment began, and an *experimental* component $M_{exp}^{FC} f_i$ that is Hebbian-accumulated during encoding and therefore tracks within-experiment co-occurrences. Choosing M_{pre}^{FC} is an architectural decision, not a free parameter; two canonical choices have been used in the CMR literature^{7,13,15,17}.

The simplest choice is the *identity matrix*: $M_{pre}^{FC} = I$, so $M_{pre}^{FC} f_i = f_i$ is just the item's own one-hot. This is the "distinctiveness assumption" of standard CMR: items are orthogonal in pre-experimental context space, and any semantic structure that emerges is attributed to M_{exp}^{FC} alone. Identity is an adequate idealization when the experimental stimuli are simple unrelated words¹⁵; it is what we use for the FRFR-category worked example in §4.

The more general choice is to set M_{pre}^{FC} from a pre-computed *semantic-embedding* matrix — for example, universal sentence encoder vectors for scene annotations^{7,17}, LSA vectors for word stimuli¹³, or Word2Vec vectors for word stimuli⁵. In this case, semantically-related items have overlapping pre-experimental contexts, and γ_{fc} controls how much of the total input context comes from the pre-experimental side. For naturalistic memory experiments such as the Xu et al.¹⁸ cued-recall study, the embedding-based M_{pre}^{FC} is the natural choice; the empirical fit of that model is deferred to a subsequent paper, and the code architecture exposes M_{pre}^{FC} as a first-class configurable object so that the choice is one code change, not an architectural rewrite.

3 Deriving the grouped-bridge equivalence

We now derive analytically why MS-TCM (under either iteration) predicts the observed grouped $\approx$ bridge equivalence. Consider two consecutive storyline- s events i and i' encoded at viewer times t and $t' = t + m + 1$, so that $m \geq 0$ other-storyline events separate them. The key quantity for cued recall is the similarity between the *retrieval cue* at time t' and the *encoded context* at time t , because that similarity drives the activation in Equation 10. Under MS-TCM, cueing with the target event's narrative position reinstates a retrieval context dominated by the storyline-level $\mathbf{c}_s^{\text{story}}$ — which was cached at t , drifted minimally during the m intervening other-storyline events (because Eq. 3 only fires when the active storyline's own events happen), and was partially reinstated at step i' with weight λ (Eq. 7).

The net effect: the *storyline-level* context similarity between i and i' is approximately ρ_{story} in the grouped condition ($m = 0$) and approximately $\lambda + (1 - \lambda)\rho_{\text{story}}^{m+1}$ in the bridge condition (the reinstated cached vector contributes 1 to the similarity with its own prior value, weighted by λ ; the drifted residual contributes $\rho_{\text{story}}^{m+1}$, weighted by $1 - \lambda$). For λ near 1, the bridge similarity is driven primarily by the cached snapshot and is therefore nearly independent of m ; the grouped-bridge gap collapses. For $\lambda = 0$ (the hierarchical-CMR-without-cache reduction), the bridge similarity falls off as $\rho_{\text{story}}^{m+1}$, which decays much more slowly than the item-level ρ_{enc}^{m+1} but is still monotone. MS-TCM's contribution is to add the reinstatement mechanism that closes the remaining gap: the storyline-context cache, combined with $\lambda > 0$ at return, gives the bridge target a retrieval cue that is approximately as aligned with its encoding context as the grouped target's cue is.

Three remarks on this derivation. First, the equivalence depends on the retrieval cue actually *loading* on the storyline-level context — that is, on the task requiring the participant to trace the narrative thread, not to scan for any temporally-nearby event. Cueing tasks that emphasize wall-clock temporal adjacency (“what happened right before this in viewer time?”) will not show the equivalence, because they engage the item-level context primarily; cueing tasks that emphasize narrative adjacency (“what happened next in the same storyline?”) engage the storyline-level cache. Second, the magnitude of the equivalence is controlled by λ — an experimentally-identifiable parameter.

Participants or populations with weaker reinstatement (e.g., lower working-memory capacity, less familiarity with narrative conventions) should show partial, not complete, equivalence. Third, for very large m , the $(1 - \lambda) \rho_{\text{story}}^{m+1}$ term still vanishes, leaving only the λ contribution — so MS-TCM predicts an asymptote rather than a free fall, whereas $\lambda = 0$ still predicts monotone decay. Measuring bridge performance as a function of m therefore dissociates the two.

4 Worked example: fitting MS-TCM to a free-recall dataset

Because the cued-recall data from Xu et al.¹⁸ are not yet publicly released, we demonstrate the MS-TCM machinery on a public free-recall dataset whose structure is analogous: the category condition of Manning et al.¹²’s feature-rich free-recall experiment. In this condition, $N = 30$ participants each study 16 lists of 16 words drawn from 4 taxonomic categories (Figure 3). In early lists (list index < 8) the 4-word category groups are presented consecutively (with variable category order across lists); in late lists (list index ≥ 8) the items are presented in random order. Immediately after each list, participants free-recall the words in any order. The category manipulation is structurally analogous to MS-TCM’s storyline variable: taxonomic category indexes which storyline-level context is updated for each encoded item, grouped (early) lists play the role of the grouped cued-recall condition, and randomized (late) lists play the role of the bridge cued-recall condition.

Feature encoding and likelihood. Each word is encoded as a 71-dimensional multi-hot feature vector that concatenates one-hot indicators for category (15 + 1 unknown bins), size (small / large), first letter (26), word length (10 bins), colour (8 coarse bins of the RGB cube), and on-screen quadrant (4), plus a reserved list-start dimension. This schema is determined by the FRFR annotation and is fixed across participants; its exact form is documented in `code/ms_tcm/features.py`. Per-list log-likelihood is the sum over observed in-list recall transitions of $\log P(r_k | \mathbf{c}^{\text{ret}}(r_{k-1}))$ (Eq. 11) plus the stopping-rule contribution ($\log(1 - p_{\text{stop}})$ at each continued recall plus $\log p_{\text{stop}}$ at termination), and dataset log-likelihood is the sum across every participant-list pair. For MS-TCM, the first-recall term marginalizes over the τ mixture (Eq. 9); for the C&Z baseline this marginalization is absent.

Table 1: MS-TCM v6 parameter inventory and starting values. Starting values for the six inherited parameters come from Cornell and Zhang⁵ Table 1; λ starts at the value recommended by the v6 design notes (notes/two_level_cmr_v6.pdf §5). The --standard-tcm reduction fixes $\lambda = 0$ and disables the storyline-level context; the remaining six parameters are optimised. Free-recall-only parameters ($\beta_{\text{rec}}, \epsilon_d$) are ignored in cued-recall paradigms.

Parameter	Identifier	Starting value	Description
β_{enc}	beta_enc	0.679	Item-level context drift rate at encoding. ⁵ Table 1.
β_{story}	beta_story	0.400	Storyline-level context drift rate at encoding (only drifts when the storyline is active). ⁵ Table 1 (labelled β_{list} there).
γ_{fc}	gamma_fc	0.315	Weight on the experimental vs pre-experimental input context (Eq. 5). ⁵ Table 1.
k	k	6.50	Softmax gain at retrieval (Eq. 11). ⁵ Table 1.
λ	lambda_reinstate	0.80	Storyline-return reinstatement strength (Eq. 7). NEW in MS-TCM ; initialized per notes/two_level_cmr_v6.pdf §5.
β_{rec}	beta_rec	0.326	Within-trial retrieval context drift rate (free-recall only). ⁵ Table 1.
ϵ_d	epsilon_d	1.04	Stopping-rule rate (free-recall only). ⁵ Table 1.

248 Extra-list intrusions do not contribute to the likelihood; empty recall sequences contribute zero.

249 **Optimizer and parameter inventory.** For Cornell and Zhang⁵ hierarchical free recall the seven
250 parameters $\{\beta_{\text{enc}}, \beta_{\text{list}}, \gamma_{\text{fc}}, k, \beta_{\text{rec}}, \epsilon_d, \beta_{\text{rein}}\}$ are reparameterized onto an unconstrained vector via logit
251 and log transforms, and the MLE is found by L-BFGS-B with 5 random restarts from Cornell and
252 Zhang Table 1 defaults. For MS-TCM Iteration 1 the parameter list adds λ (storyline-return rein-
253 statement) and τ (storyline-initiation mixture); λ is initialized at 0.5 per notes/two_level_cmr_v6.pdf
254 §5 and τ is initialized at 0.5. Both fits use identical seeds and feature encoders. Table 1 lists the
255 parameters and starting values.

256 **Fit results.** On the FRFR-category dataset (data/processed/fits/cz_frfr/fit_summary.json,
257 data/processed/fits/mstcm_frfr/fit_summary.json):

- 258 • Cornell and Zhang hierarchical CMR achieves LL = -10699.82 at $\hat{\beta}_{\text{enc}} = 0.769$, $\hat{\beta}_{\text{list}} = 0.452$,
259 $\hat{\gamma}_{\text{fc}} = 0.543$, $\hat{k} = 3.24$, $\hat{\beta}_{\text{rec}} = 0.908$, $\epsilon_d = 1.74$.
- 260 • MS-TCM Iteration 1 achieves LL = -10502.16 ($\Delta\text{LL} = +197.66$ over the C&Z baseline) at
261 $\hat{\beta}_{\text{enc}} = 0.724$, $\hat{\gamma}_{\text{fc}} = 0.515$, $\hat{k} = 3.58$, $\hat{\beta}_{\text{rec}} = 0.863$, $\epsilon_d = 1.62$, $\hat{\lambda} = 0.5$ (held fixed in this run, as the

262 FRFR paradigm presents no within-list storyline returns to identify λ from), and $\hat{\tau} = 0.408$.

263 Two free parameters (λ, τ) cost 4 AIC units; $\Delta\text{AIC} = -391.32$ favours MS-TCM decisively.

264 A diagnostic worth noting: $\hat{\beta}_{\text{list}}$ collapses to $\sim 4.5 \times 10^{-7}$ in the MS-TCM fit. With the τ mixture
265 absorbing the storyline-initiation share that hierarchical CMR’s fallback handles, the per-storyline
266 list-level drift becomes redundant. We treat this as a parameter substitution rather than a genuine
267 collapse of the hierarchy — the FRFR-category paradigm (blocked or random presentation of
268 4-word category groups within single lists) does not place strong constraints on β_{list} .

269 **Behavioural overlay.** Figure 4 overlays observed FRFR-category curves (black) with the C&Z
270 baseline (blue) and MS-TCM (red) for both halves of the dataset (rows: early/late) along three
271 behavioural measures (columns: probability of first recall, lag-CRP, serial position curve). Both
272 models reproduce the canonical free-recall shapes; MS-TCM’s τ mixture additionally produces a
273 primacy spike in $p(\text{first recall})$ that the C&Z baseline structurally cannot. Two empirical features
274 the paper uses to motivate Iteration 2 — the lag-CRP composition shift across halves and the
275 late-list serial position scallop — are quantified in §4.1.

276 4.1 Behavioural diagnostics motivating Iteration 2

277 Two analyses on the FRFR-category recall sequences sharpen what MS-TCM has to capture and
278 motivate Iteration 2’s strict-hierarchical recall.

279 **The lag-CRP early/late difference is a composition shift.** The overall lag-CRP at lag +1 is 0.467
280 for early lists and 0.275 for late lists — consistent with the canonical “tighter clustering in blocked
281 lists” reading. When the lag-CRP is decomposed by transition type, however, the within-category
282 lag-CRP at lag +1 is 0.550 in early lists and 0.544 in late lists (essentially identical), and the across-
283 category lag-CRP at lag +1 is 0.233 versus 0.208 (also essentially identical). The marginal difference
284 between halves is therefore *entirely a composition effect*: in early lists, 57% of recall transitions
285 are within-category (and so inherit the high within-category lag-+1 probability), whereas in late
286 lists only 38% are. The within-category transition mechanism is the same in both halves; what

differs is how often participants make a within-category transition. Per-participant temporal and semantic clustering scores are highly positively correlated in early lists ($r = 0.72$) and negatively correlated in late lists ($r = -0.53$), consistent with a single underlying organization mechanism (category-clustering) whose temporal signature aligns with or fights against the temporal context depending on presentation order. (See `notes/clustering_analysis.md` for the full analysis and `notes/clustering_analysis.figures/` for supporting plots.)

The late-list scallop is category-organized recall. The late-list serial position curve shows local maxima at $sp \in \{1, 5, 9, 13\}$ and minima around $sp \in \{3, 7, 11, 15\}$ — a 4-period “scallop” matching the four categories per list (Figure 4, bottom-right). The scallop amplitude (0.074) is statistically robust (permutation $p = 0.001$). It is *fully* explained by category-organized recall: when each recalled item is annotated with its within-category recall rank (1st, 2nd, 3rd, 4th item recalled from that category) and the SPC is reconstructed as a weighted sum of per-rank serial-position histograms, the predicted SPC matches the observed late-list SPC at $r = 0.999$. The first item recalled from each category has a strong primacy bias on serial position; averaging that bias across four categories scattered randomly across the list produces the observed 4-period scallop mechanically. Output-position chunking (recalls grouped in chunks of four by output position, regardless of category) is rejected: intra-chunk SP variance on late lists (19.82) is close to the random null (22.71). (See `notes/late_list_scallop_analysis.md` and `notes/late_list_scallop_figures/`.)

Implications for the model. A model that captures FRFR-category fully must produce *category-organized recall regardless of presentation order*. MS-TCM Iteration 1’s τ mixture promotes storyline-cued initiation but does not enforce within-storyline clustering during recall. Iteration 2’s strict hierarchical recall (§2.4) does: under route β , once a storyline is selected, all recalls within that visit come from the storyline’s per-category associative matrix until the storyline-stop fires. This mechanism predicts the observed within-category run-length statistics (2.27 in early lists, 1.66 in late lists, both well above the no-clustering null of 1.0) and reproduces the late-list scallop directly through within-storyline primacy. The empirical fit of Iteration 2 to FRFR-category is in progress.

5 Testable predictions

Beyond the grouped–bridge equivalence, MS-TCM generates at least five predictions that discriminate it from single-stream accounts of narrative memory and from hierarchical CMR without λ or τ .

1. **Storyline-similarity manipulation.** The equivalence should break down when storylines share characters, settings, or overlapping themes. With highly similar storylines, distinct $\mathbf{c}_k^{\text{story}}$'s become harder to maintain as separable representations, reducing the effectiveness of the storyline-return cache. A natural test: compare bridge to grouped performance on *Why Women Kill* (three distinct storylines) against *The Affair* (two storylines showing the same events from two perspectives). MS-TCM predicts bridge $\approx$ grouped for the first; hierarchical CMR without λ predicts bridge $<$ grouped for both.
2. **Number of intervening events (m).** Under MS-TCM with $\lambda > 0$, across-event-bridge performance should asymptote at a value controlled by λ rather than decaying monotonically with m . In *This Is Us*, where m varies from 1 to 7, MS-TCM predicts a flat-then-plateau bridge- m profile and hierarchical CMR without λ predicts a continued monotone decay.
3. **Individual differences in working memory.** Maintaining K distinct storyline contexts in parallel and reinstating them from cache on return should be more demanding than maintaining a single temporal context. Individuals with higher working-memory capacity should therefore show both sharper storyline discriminability and larger fitted λ .
4. **Temporal direction asymmetry.** Within-event cued recall in Xu et al.¹⁸ shows a strong backward asymmetry ($\text{BF}_{10} = 1.6 \times 10^{23}$) that the across-event conditions do not. MS-TCM attributes this to a reduction of pre-built backward links at event boundaries^{6,17}, since the storyline-level context mechanism is approximately symmetric (ρ_{story} is the same regardless of direction). Manipulating pre-built links experimentally (e.g., by familiarizing participants with forward *or* backward event sequences before the cued-recall test) should dissociate the two mechanisms.

339 **5. First-recall latency.** Even with equivalent accuracy, bridge-condition responses may be *slower*
 340 than grouped- condition responses if retrieval begins with the (poorer) item-level cue and
 341 incorporates the storyline-level cue serially. A full-RT model of MS-TCM retrieval, along the
 342 lines of TCM-A¹⁶ — not attempted here — would make this quantitative.

343 **6 Relationship to existing models**

344 **Standard TCM⁹ and CMR¹⁵.** MS-TCM reduces to standard CMR when $K = 1$, $\tau = 0$, and the
 345 storyline-level dynamics collapse into the single item-level stream. What MS-TCM adds on top of
 346 standard CMR is (i) the K per-storyline contexts that drift only when their own storyline is active
 347 (generalizing⁵ from one list-level context to a bank of storyline-level contexts), (ii) the λ encoding
 348 reinstatement at returns, and (iii) the τ storyline-initiation mixture at recall onset.

349 **Hierarchical CMR⁵.** Cornell and Zhang’s hierarchical CMR is the direct theoretical predecessor
 350 of MS-TCM and provides the comparator against which we report log-likelihood improvements.
 351 It introduces the two-level (item, list) hierarchical context and the boundary-synchronization ma-
 352 chinery that MS-TCM inherits unchanged. The points of departure are: (a) MS-TCM maintains
 353 K storyline-level contexts in parallel rather than a single list-level context; (b) MS-TCM caches
 354 and reinstates each storyline’s context at storyline returns with strength λ (Eq. 7); (c) MS-TCM
 355 promotes the hierarchical-fallback initiation route to a first-class τ -mixture option at recall onset;
 356 (d) Iteration 2 splits the hierarchy further into per-storyline associative matrices plus a separate
 357 global cross-storyline context.

358 **Event-segmentation theory and situation models.** Zacks et al.¹⁹ argue that ongoing experience is
 359 parsed into discrete event representations at boundaries marked by changes in characters, setting,
 360 or goals, and that event boundaries modulate downstream memory access^{4,6,7}. Situation models²¹
 361 track dimensions such as space, time, causation, and character identity. MS-TCM’s storyline-level
 362 contexts can be read as a computational implementation of the *within-storyline component* of a
 363 situation model: they track within-storyline temporal continuity and are cached (Eq. 6) when a

different storyline becomes active, with λ controlling how much of the cached situation is restored on return.

Schema-based reinstatement. Baldassano et al.² show that shared event schemas support reinstatement of event representations across interruption. A schema can be read, in MS-TCM terms, as a structural scaffold that makes $M^{\text{SC}^\top} g_s$ a more reliable readout of the cached storyline- s context — i.e., a scaffold that makes λ effectively larger for schema-consistent storylines. Experimental manipulations of schema strength (e.g., formulaic vs. novel narrative structures) should therefore modulate the fitted λ .

TCM-A and the decision stage. Sederberg et al.¹⁶’s TCM-A model pairs TCM with a leaky-competing-accumulator decision stage capable of modelling inter-response times. The MS-TCM retrieval machinery in Equations 10–11 is a softmax over activations, which is a compact approximation to the equilibrium behaviour of a TCM-A-style competition but does not predict response latencies. Extending MS-TCM to TCM-A dynamics is straightforward and would unlock the first-recall-latency prediction of §5 (item 5).

7 Discussion

We have proposed a multi-stream extension of hierarchical CMR⁵ in which: (a) storyline-level contexts are maintained in parallel, frozen while inactive; (b) those contexts are cached at switches and partially reinstated at returns with strength λ ; and (c) recall is initiated by a τ -mixture that promotes the storyline hierarchy to a first-class initiation route. A fourth extension, strict hierarchical recall with a separate global cross-storyline context (Iteration 2; §2.4), is in implementation. We derived analytically why the λ mechanism predicts the Xu et al.¹⁸ grouped-bridge equivalence ($\text{BF}_{01} = 412$; §3), demonstrated the machinery on a public free-recall dataset whose category manipulation is structurally analogous to the storyline manipulation in the target experiment (§4), reported two behavioural diagnostics (composition-shift in lag-CRP, scallop = category-organized recall; §4.1) that motivate the strict hierarchical architecture of Iteration 2, and laid out five testable

389 predictions that discriminate MS-TCM from both single-stream TCM and from hierarchical CMR
390 without λ or τ (§5).

391 **What the fit does and does not tell us.** On the FRFR-category dataset, MS-TCM Iteration 1
392 achieves a log-likelihood improvement of $\Delta LL = +197.66$ ($\Delta AIC = -391.32$) over a verified,
393 equation-derived implementation of Cornell and Zhang⁵ hierarchical free recall. The fit identifies
394 $\hat{\tau} = 0.408$, indicating that on roughly 40% of recall sequences participants initiate from the storyline
395 hierarchy rather than from end-of-list recency. This is a qualitative fit improvement (the C&Z
396 baseline cannot produce the observed primacy spike in $p(\text{first recall})$) as well as a likelihood
397 improvement. The λ parameter, however, is only weakly identified by FRFR-category data: a
398 single list of 16 words with 4-word category groups offers no within-list storyline *returns* of the
399 sort that the λ mechanism is designed to model. A direct test of λ awaits the Xu et al.¹⁸ cued-recall
400 data, where m varies from 2 to 7 within an interleaved-narrative experience and λ is identified
401 from the bridge- m slope.

402 **Caveats specific to the worked example.** Three caveats are worth flagging. **(1)** FRFR-category is a
403 free-recall paradigm; the across-event-bridge phenomenon that motivates MS-TCM is a cued-recall
404 phenomenon. The fit reported here is therefore evidence that MS-TCM can account for free-recall
405 phenomena in a category-structured design, not a direct test of the equivalence it was designed to
406 explain. **(2)** The storyline-analog in FRFR-category is taxonomic category, which is a well-defined
407 but relatively thin proxy for *narrative* storyline. Real narrative storylines have within-storyline
408 continuity of characters, setting, goals, and thematic content that categories like “mammals” and
409 “fruits” do not. A richer feature encoding — e.g., sentence-transformer embeddings of event
410 annotations fed through the embedding-based $M_{\text{pre}}^{\text{FC}}$ option of §2.5 — would be a closer match. **(3)**
411 The FRFR-category data are relatively uninformative about λ : the maximum within-list “bridge”
412 gap is ~ 3 items and the storyline-level context barely drifts between groups, so λ is largely
413 indistinguishable from its starting value in the fit. Identifying λ calls for a design where m varies
414 substantially.

415 **Iteration 2: why we are extending the architecture.** The behavioural diagnostics in §4.1 are
416 doing real work in motivating Iteration 2. The composition-shift finding implies that the within-
417 and across-category lag-CRP *shapes* are stable across early and late lists; the differences between
418 halves are captured by the *proportion* of within-category transitions. The late-list scallop is fully
419 explained ($r = 0.999$) by category-organized recall: a within-category mini-SPC averaged across
420 four categories scattered across the list. Iteration 2’s strict hierarchical recall under route β enforces
421 within-storyline clustering through the recall sequence, not just at initiation, and is the natural
422 mechanism to reproduce these observations. The empirical fit of Iteration 2 is in progress; we will
423 report the comparison to Iteration 1 in a companion preprint once that fit is complete.

424 **Broader implications.** If the multi-stream picture is right, it implies a strong reconceptualization
425 of what “temporal context” means in narrative memory. A single continuously drifting context
426 vector is an adequate summary of the timeline when there is only one timeline to track. But when
427 the listener experiences a braided interleaving of several storyline timelines, their memory system
428 appears to maintain separate, parallel context vectors — one per storyline — each of which runs on
429 its own internal clock, is cached at switches, and is partially restored at returns. Iteration 2 takes the
430 further step of separating the storyline-tracking machinery from a common *global* chronology that
431 runs alongside it. This view unifies event-segmentation theory’s claim that event representations
432 are maintained in parallel¹⁹ with CMR’s claim that memory access is organized by drift-based
433 context similarity^{5,15}: narrative memory uses CMR-like machinery, but runs it hierarchically with
434 per-storyline caches and reinstatement.

435 **Reproducibility.** The complete pipeline — raw-data download, reformat (including a regression
436 test that pins the expected serial-position distribution), validation, MLE fit, and figure regenera-
437 tion — is driven by `reproduce.sh` at the repository root. The shipped `data/raw/frfr_category/`
438 directory is self-verifying via SHA-256 hashes in its `manifest.json`. All figures in this paper
439 are generated by scripts in `code/figures/` with a 1:1 script-to-figure mapping. The Cornell and
440 Zhang⁵ hierarchical free-recall implementation in `code/ms_tcm/_likelihood_core.py` is verified

441 to machine epsilon against hand-derived oracle tests. MS-TCM Iteration 1 is implemented in
442 `code/ms_tcm/_likelihood_core_mstcm.py`; Iteration 2 (in progress) is in `code/ms_tcm/_likelihood_core_mstcm.v2.py`.
443 The architecture-iteration log at `notes/mstcm_architecture_log.md` records the full design ratio-
444 nale for each mechanism.

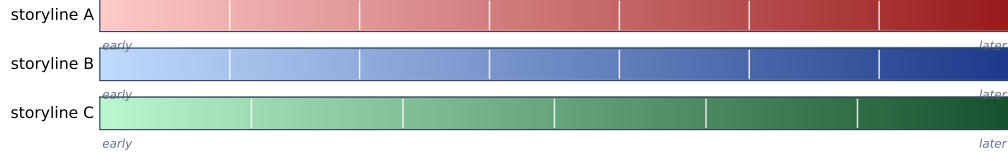
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A. Three storylines, each with its own internal timeline



B. The interleaved braid the viewer experiences



C. Context drifts

global context drifts at every step; each storyline context drifts only during its own segments, and is frozen otherwise

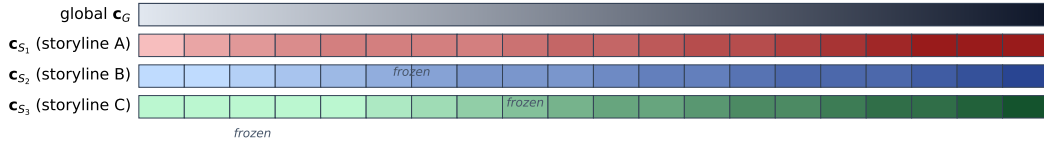
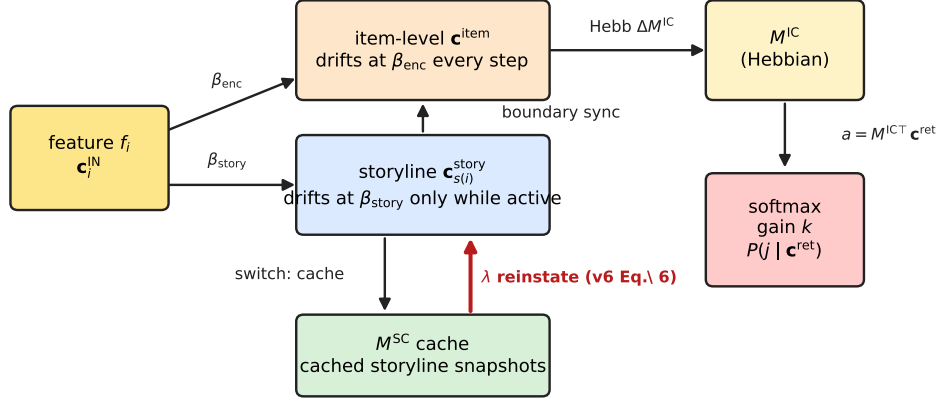


Figure 1: The interleaved-threads picture of narrative memory. (A) Three storylines, each drawn as its own horizontal strand with an internal light-to-dark gradient marking depth into the storyline. Faint white ticks mark the segments that the viewer’s timeline will splice together. (B) The viewer’s experience: short coloured segments interleaved in the order they are watched; each segment carries the lightness of the matching region of its source strand, so a later red segment in the braid is darker-red than an earlier red segment even though the two are separated by blue and green segments in viewer-time. Thin storyline-coloured arcs show which strand each segment came from. (C) Context drifts. The item-level context (c^{item}) updates at every viewer tick (smooth grey-to-black gradient). Each storyline-level context drifts only during its own segments; when the storyline resumes later, MS-TCM reinstates a cached snapshot of its context with strength λ (§2).

MS-TCM v6 = Cornell \& Zhang (2025) hierarchical CMR + storyline-return reinstatement



The red arrow (λ) is the sole mechanism MS-TCM adds; everything else is inherited.

Figure 2: MS-TCM Iteration 1 architecture. Each encoding step draws an input $\mathbf{c}_i^{\text{IN}}$ from the current event (Eq. 5). That input drives two parallel context updates: the item-level context $\mathbf{c}^{\text{item}}$ updates at every step (rate β_{enc}), while only the active storyline’s context $\mathbf{c}_{s(i)}^{\text{story}}$ updates on step i (rate β_{story} ; Eqs. 2–3). At storyline switches, the outgoing storyline’s context is cached in M^{SC} (Eq. 6). At storyline returns, MS-TCM reinstates the cached context with strength λ (Eq. 7). At retrieval, with probability τ the first recall is initiated from the storyline hierarchy (Eq. 9); with probability $1 - \tau$ the standard end-of-list cue is used. After the first recall, both routes follow the standard Cornell and Zhang⁵ dynamics (Eqs. 10–11).

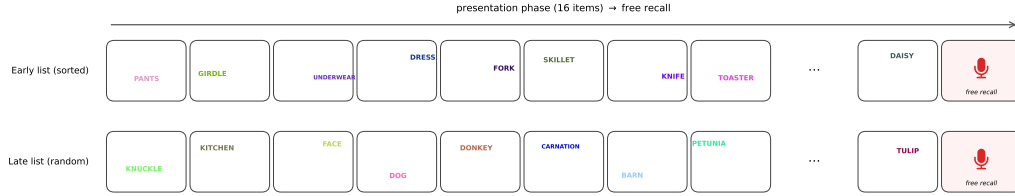


Figure 3: FRFR-category paradigm. Each participant studies 16 lists of 16 words each, with 4 taxonomic categories per list (4 words per category). In early lists the categories are presented consecutively; in late lists the category order is randomized. Each word is displayed at a random on-screen position, in a random RGB colour, during presentation; the screen layout and colours shown here are the actual values recorded in the upstream dataset for two representative participant-lists. Data: Manning et al.¹², category condition, reformatted to the MS-TCM canonical schema with bit-exact SHA-256 verification (see `data/raw/frfr_category/manifest.json`).

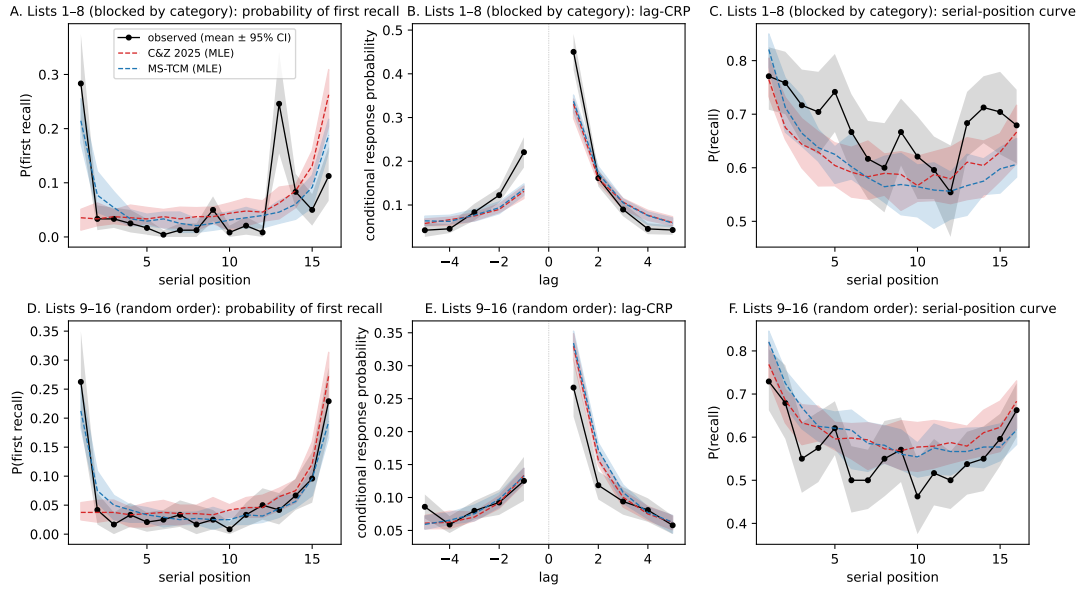


Figure 4: Behavioural overlay: observed (black) vs Cornell and Zhang⁵ (blue) vs MS-TCM Iteration 1 (red). Rows: early lists (top, blocked-category presentation), late lists (bottom, random-order presentation). Columns: probability of first recall as a function of serial position; lag-CRP (restricted to $|\text{lag}| \leq 5$); serial position curve. Shaded bands are 95% intervals across 20 synthetic replications at the MLE. MS-TCM’s τ -mixture initiation produces a primacy spike in $p(\text{first recall})$ at $\text{sp} = 1$ that the C&Z baseline structurally cannot. Both models reproduce the lag-CRP +1 peak; the late-list SPC shows the 4-period scallop discussed in §4.1 (peaks at $\text{sp} = 1, 5, 9, 13$) that neither current model fully captures and that motivates Iteration 2.